

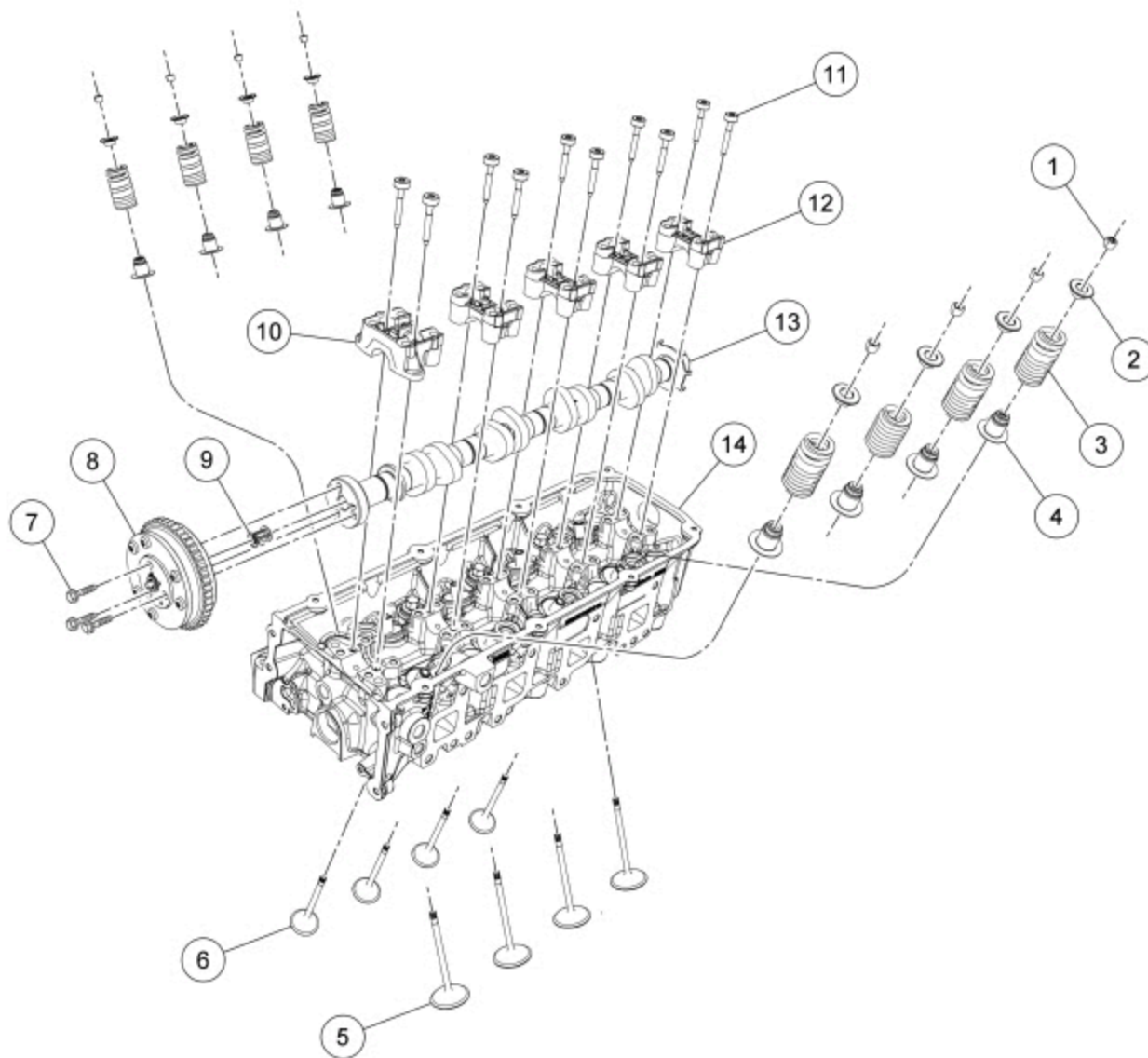
Cylinder Head



Material	
Item	Specification
Motorcraft® SAE 5W-20 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)	WSS-M2C945-A

Intake and Exhaust Valves, Valve Springs, Valve Spring Retainer Keys and Retainers, Camshaft Phaser and Sprocket, VCT System Oil Filter, Camshaft Bearing Caps and Camshaft

NOTE: *LH shown, RH similar.*



N0110700

Item	Part Number	Description
1	6518 6518	Valve spring retainer key (16 required)
2	6514 6514	Valve spring retainer (8 required)
3	6513 6513	Valve spring (8 required)
4	6A517 6A517	Valve seal (8 required)
5	6505 6505	Exhaust valve (4 required)
6	6507 6507	Intake valve (4 required)
7	W712998 W712998	Camshaft phaser and sprocket bolt (3 required)
8	6C524 6C524	Camshaft phaser and sprocket
9	6C683 6C683	Variable Camshaft Timing (VCT) system oil filter
10	6B280 6B280	Camshaft front bearing cap
11	W714203 W714203	Camshaft bearing cap bolt (10 required)
12	6B284 6B284	Camshaft bearing cap (4 required)
13	6A274 6A274	Camshaft
14	6050 6050	Cylinder head

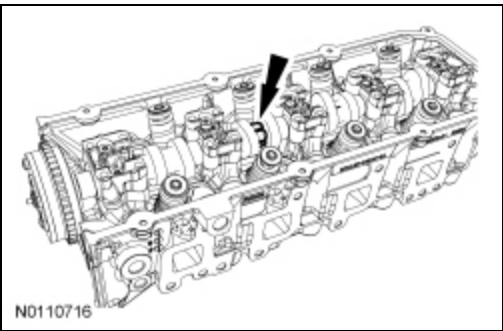
Disassembly

1. **NOTICE:** Damage to the camshaft phaser and sprocket assembly will occur if mishandled or used as a lifting or leveraging device.

NOTICE: Only use hand tools to remove the camshaft phaser and sprocket assembly or damage may occur to the camshaft or camshaft phaser and sprocket.

Using a 26 mm (1.023 in) wrench on the flats of the camshaft to secure the camshaft, remove the 3 RH camshaft phaser and sprocket bolts and the camshaft phaser and sprocket.

- Discard the 3 camshaft phaser and sprocket bolts.



2. Remove and discard the Variable Camshaft Timing (VCT) system oil filter.
3. **NOTICE:** If the components are to be reinstalled, they must be installed into their original locations. Failure to follow this instruction may result in engine damage.

Remove the 10 bolts and the 5 camshaft bearing caps.

4. Remove the camshaft.
5. **NOTICE:** If the components are to be reinstalled, they must be installed into their original locations. Failure to follow this instruction may result in engine damage.

Using a valve spring compressor, remove the valve spring retainer keys, valve spring retainer, the valve spring and the valve seal.

- Discard the valve seal.

6. Remove the valve from the cylinder head.
7. Repeat the previous 2 steps for each valve.
8. Inspect the components. For additional information, refer to Section 303-00.
9. Check the cylinder head for distortion. For additional information, refer to Section 303-00.

Assembly

All cylinder heads

1. **NOTE:** Lubricate the valve stem with clean engine oil prior to installation.

Install the valve into the cylinder head.

2. **NOTE:** Lubricate the valve seal and valve stem with clean engine oil prior to installation.

Position a new valve seal onto the valve stem.

3. Using a valve stem oil seal installer, install the new valve seal.

4. **NOTICE: If the components are to be reinstalled, they must be installed into their original locations. Failure to follow this instruction may result in engine damage.**

Using a valve spring compressor, install the valve spring, the valve spring retainer and the valve spring retainer keys.

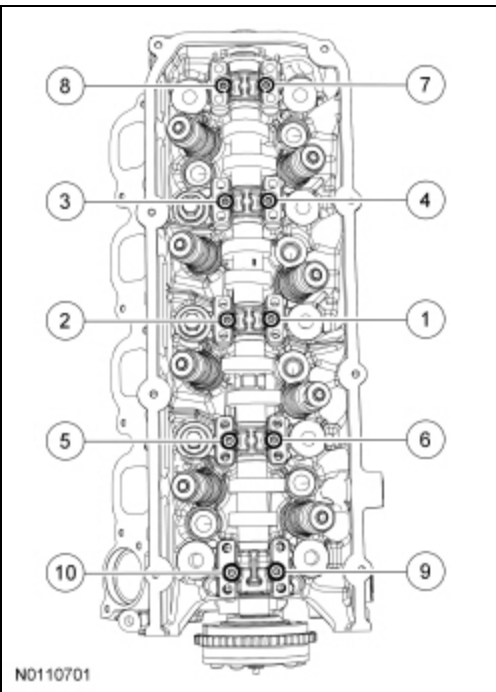
5. Repeat the previous 4 steps for each valve.
6. Lubricate the camshaft and camshaft journals with clean engine oil and install the camshaft.

LH cylinder head

7. **NOTICE: If the components are to be reinstalled, they must be installed into their original locations. Failure to follow this instruction may result in engine damage.**

Position the camshaft bearing caps in their original locations and install the 10 bolts in the sequence shown in 2 stages.

- Stage 1: Tighten to 6 Nm (53 lb-in).
- Stage 2: Tighten an additional 45 degrees.

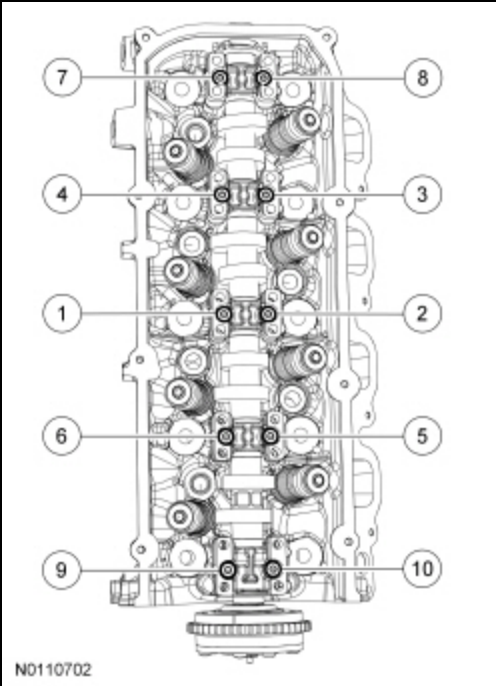


RH cylinder head

8. **NOTICE: If the components are to be reinstalled, they must be installed into their original locations. Failure to follow this instruction may result in engine damage.**

Position the camshaft bearing caps in their original locations and install the 10 bolts in the sequence shown in 2 stages.

- Stage 1: Tighten to 6 Nm (53 lb-in).
- Stage 2: Tighten an additional 45 degrees.



All cylinder heads

9. Install the new VCT system oil filter into the camshaft, with the open end facing the front of the engine.

10. **NOTICE: Damage to the camshaft phaser and sprocket assembly will occur if mishandled or used as a lifting or leveraging device.**

Using a 26 mm (1.023 in) wrench on the flats of the camshaft to secure the camshaft, tighten the 3 new LH camshaft phaser and sprocket bolts in 2 stages:

- Stage 1: Tighten to 9 Nm (80 lb-in).
- Stage 2: Tighten an additional 90 degrees.

